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Effect of Supplementing Herbal Feed Additive *Anethum sowa* on Nutrient Utilization, Productive Performance and Carcass Characteristics of Male Beetal Kids

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ABSTRACT

Hundal, J.S., Wadhwa, M. and Bakshi, M.P.S. 2020. Effect of supplementing herbal feed additive *Anethum sowa* on nutrient utilization, productive performance and carcass characteristics of male beetal kids. *Animal Nutrition and Feed Technology*, 20: 25-38.

A 90d growth trial was conducted on eight male Beetal goat kids (5 months old; average BW 14.85 ± 0.31 kg) to assess the effect of supplementing *Anethum sowa*- a herbal feed additive (HFA) containing essential oils - on their nutrient utilization, productive performance and meat quality. The animals, divided into two equal groups, were offered either a control total mixed ration (CON) or the same supplemented with *Anethum sowa* at 2.0% of DM intake (HFA). Rumen studies were conducted simultaneously on three adult Beetal bucks fitted with permanent rumen fistulae adopting a complete crossover design involving two periods. Results indicated that the concentrations of various N fractions were higher ($P < 0.01$) in the rumen contents of animals in the HFA group as compared to the CON; however, the ammonia-N as percent of total-N was depressed ($P < 0.01$) in the HFA group. The total and individual VFAs concentrations were improved ($P < 0.01$) in HFA group, with improved acetate to propionate ratio. The daily DM intake, digestibility of nutrients, N-retention, blood profile and urinary purine derivatives, and ADG were statistically comparable in both the groups. The dressing percentage and edible offal percentage were similar between the two groups. The ready-to-cook carcass expressed on weight basis ($P < 0.05$) or as percent of live weight ($P < 0.01$) basis was improved significantly in HFA group. The weight of non-carcass components was improved in the group fed HFA as compared to CON group. It was concluded that total mixed ration supplemented with *A. sowa* at 2.0% of DM intake improved the carcass characteristics of Beetal goat kids.

Keywords: Beetal kids, Carcass quality, Essential oil, Nutrient utilization, Purine derivatives, Rumen metabolites

INTRODUCTION

India has a goat population of 134 million with a goat meat production of 1.04 million tonnes/annum (FAO, 2016). The annual growth rate of goat meat production is 2.4%. India exports only 2-3% of the total meat produced whereas the rest is

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consumed locally (Anonymous, 2018). Goats are largely reared by landless, small and marginal farmers and fed mainly on tree leaves/crop residues based diets (Bakshi and Wadhwa, 2007). There is great scope of increasing growth rate, average daily gain in weight and margin of profit by scientific feeding under field conditions. Further, efforts are afoot to improve the carcass traits by supplementing the balanced ration with phytogenic feed additives. Earlier studies conducted by using pure essential oils (Hundal *et al.*, 2016a) or tannins (Hundal *et al.*, 2016b) revealed significant reductions in *in vitro* methane production from the substrate with or without affecting digestibility of nutrients and VFA production. But using pure essential oils may not be easy to handle by the farmers under field conditions. Therefore, use of dry extract (as fine powder) of herbal feed additives (HFA) rich in plant secondary metabolites (PSMs) are easy to handle and may have synergistic effect on the rumen fermentation pattern and methane mitigation.

In an earlier *in vitro* study wherein six HFAs containing essential oils were supplemented to total mixed rations (TMR), it was revealed that the dry extract of *Anethum sowa* seeds when supplemented at 2% of the TMR was effective in improving the total and individual VFA production, reduced methane and ammonia production as compared to un-supplemented control TMR (Hundal *et al.*, 2019). The dry extract of seeds of *A. sowa* contained 2.46% essential oils, 4.49% saponins, 0.15% condensed tannins and 2.27% 2, 2-diphenyl-1-picrylhydrazyl-hydrate (DPPH) on DM basis (Hundal *et al.*, 2019). However, no information is available on the effect of supplementing *A. sowa* on the productive performance of goats. The present study was, therefore, taken up to assess the effect of supplementing the TMR with *A. sowa* at the rate of 2% level on the digestibility of nutrients, rumen metabolites and carcass quality of male Beetal kids.

MATERIALS AND METHODS

A 90 d growth trial was conducted on 5 months old, male Beetal goat kids to assess the effect of supplementing selected herbal feed additive to the TMR, on nutrient utilization, productive performance and meat quality of kids. The dry extract of seeds of *Anethum sowa* as fine powder was procured from Konark Herbs and Healthcare (Mumbai, India).

Selection, distribution and maintenance of animals

Eight male Beetal goat kids (5 months old; average BW 14.85 ± 0.31 kg) were randomly distributed into two groups of four animals each. All the animals were fed a TMR as per NRC (2007) specifications. The TMR contained roughage (available green fodder) and concentrate mixture in 50:50 ratio on DM basis. The concentrate mixture contained maize 32, barley 20, soybean meal 15, groundnut cake 15, rice bran 15, mineral mixture 2 and common salt 1% each. The animals in control group (CON) were fed the TMR alone while animals in experimental group (HFA) were fed the TMR supplemented with *A. sowa* dry seed extracts at 2% of DM intake. The

required quantity of dry extract of seeds of *A. sowa* was mixed daily in TMR before feeding. The daily record of feed intake and orts was maintained. The kids were weighed for 3 consecutive days every fortnight and the feeding schedule was revised accordingly. The animals were housed in a concrete shed and were stall fed individually at 9:00 AM daily. The animals had free access to water twice a day and were taken out in the yard for exercise for 1h daily.

Metabolism trial

Before termination of 90 d growth experiment, a 7-d metabolism trial was conducted. During metabolic trial, the animals were kept individually in specially designed metabolic cages where urine and faeces were collected automatically in separate containers. For urine collection, a narrow mouth plastic container (2.5 lt. capacity) containing 30 ml of 20% H_2SO_4 was used. The faeces, urine and orts (if any) were weighed daily at 9.00 AM, before feeding. Samples of feedstuffs and orts were collected at 24 h interval and dried in duplicate at 80°C in a hot air oven. After thorough mixing, a 20% sub-sample of the faecal void of each animal was taken for DM determination at 100°C in a hot air oven. For nitrogen estimation, 10% of fresh faeces samples were preserved in previously tarred wide mouth plastic bottles daily, containing 25 ml of 20% sulphuric acid. The samples of feeds and faeces were pooled for 7 d of the trial, finely ground to pass through 1-mm sieve and analyzed for proximate and cell wall constituents. An aliquot of urine equal to 10% of the total urine voided daily was preserved in narrow mouth glass bottles (500-ml capacity) and kept in a refrigerator till analyzed for nitrogen (AOAC, 2007). The urine samples were diluted 5-times with distilled water and stored at - 20°C till analyzed for purine derivatives.

Rumen fermentation study

Three adult bucks fitted with permanent rumen fistulae were used in a complete crossover design involving two periods to assess the effects of the two diets (CON and HFA) on the rumen fermentation. Each period consisted of 30 d of adaptation to the respective diets followed by collection of rumen contents at 0, 2, 4, 6, 8, 12h post-feeding for three consecutive days. Equal quantity of strained rumen liquor (SRL), from each collection, was pooled for the respective animal and preserved with a few drops of $HgCl_2$. The samples were subsequently analyzed for total nitrogen, ammonia-N (AOAC, 2007), non-protein nitrogen (NPN), trichloroacetic acid-precipitable nitrogen (TCA-N; Cline *et al.*, 1958) and VFA by GLC (Cottyn and Boucque, 1968).

Blood profile

On the last day of metabolism trial, separate jugular blood samples from each animal were collected (4 h post-feeding) into heparin and sodium fluoride-oxalate vials. The separated serum samples were stored at 0°C till analyzed.

Assessment of meat characteristics

On the last day of feeding trial, goat kids were slaughtered as per standard procedure following animal welfare guidelines in the experimental slaughter house of Department of Livestock Products Technology. The feeding was withheld 12 h prior to slaughtering. However, they were given free access to the drinking water. All the animals were humanely slaughtered using Seydelman double pronged electrical stunner with 90V-120V current for 15 sec.

The primal cuts were made by following the standard procedure (Sharma, 1999). The carcasses were divided into 8 primal cuts; neck, foreshank, shoulder, rack, brisket, breast, loin and flank and leg. After removal of all separable connective tissue, fat, fascia and blood vessels, the meat was packed in colourless low density polythene (LDPE) bags and stored in a deep freezer at - 20°C till analyzed.

Colour profile of the carcass was measured using a chromameter (CR-400, Konica Minolta, Japan) set at 2° of cool white light (D65) and known as 'L*', *a**, and *b** values. 'L*' value denotes (brightness 100) or lightness (0), '*a*' + redness/-greenness, and '*b*' + yellowness/-blueness) using the procedure as described by Kumbhar *et al.* (2018).

Analytical methods

The finely ground samples of feedstuffs (in triplicate), orts and faeces were analyzed for DM, CP and total ash (AOAC, 2007), cellulose (Crampton and Maynard, 1938) and other cell wall constituents like NDF, ADF, and ADL (Van Soest *et al.*, 1991). Hemi-cellulose was determined by difference between NDF and ADF. The urine samples were analyzed for total-N (AOAC, 2007), purine derivatives (PD), allantoin (Young and Conway, 1942) and uric acid (Trivedi *et al.*, 1978), creatinine (Folin and Wu, 1919) and microbial protein synthesis (IAEA-TECDOC, 1997). Serum samples were analyzed for albumin, (Doumas *et al.*, 1971), urea (Evans, 1968), triglycerides (McGowan *et al.*, 1983) and total cholesterol (Allain *et al.*, 1974) using Erba (Mannheim) Chem 5X (Transasia). The serum collected with sodium fluoride and oxalate was used for assay of blood glucose (Trinder, 1969).

The data were analyzed by using one way ANOVA (Snedecor and Cochran, 1994) by using SPSS (2009) version 16.0 and the means were tested for the significant difference by using Tukey's B test.

RESULTS AND DISCUSSION

The TMR fed to goat kids/bucks contained 90.66% OM, 17.5% CP, 3.45% EE, 34% NDF, 23.18% ADF, 17.11% cellulose and 11.32% hemicellulose on DM basis.

Rumen fermentation

As compared to CON, the NH₃-N, NPN, TCA-N as well as the total-N contents in rumen liquor were increased (*P* < 0.01) in HFA group (Table 1). Ammonia-

Table 1. Effect of supplementation of *A. sowa* on the biochemical changes in the rumen of Beetal bucks

Parameters [‡]	Dietary groups [†]		Pooled SE	P value
	CON	HFA		
NH ₃ -N	0.033 ^a	0.054 ^b	0.002	<0.001
TCA-N	0.12 ^a	0.24 ^b	0.02	<0.001
NPN	0.07 ^a	0.12 ^b	0.01	<0.001
Total N	0.19 ^a	0.36 ^b	0.02	<0.001
<i>N fractions as percentage of total-N</i>				
NH ₃ -N	17.37 ^b	15.00 ^a	0.45	0.001
NPN	36.84	33.33	1.93	0.147
TCA-N	63.16	66.67	1.93	0.147

[†]Feeding of basal diet in form of total-mixed ration alone (CON) or supplemented with *A. sowa* at 2% level (HFA)

[‡]TCA-N: trichloroacetic acid-precipitable nitrogen, NPN: Non-protein nitrogen

^{ab}Means with different superscripts in a row differ significantly (P<0.05)

N concentration is a balance between degradation of feed protein and uptake of ammonia for synthesis of microbial protein. Increased NH₃-N concentrations could be due to increased CP degradability (Getachew *et al.*, 2000) and/or poor synchronization between N and carbohydrate release in the rumen. The rapid release of N that is not matched to availability of carbohydrate can lead to accumulation of ammonia N *in vitro* or to high absorption of ammonia N from the rumen *in vivo*. The results of rumen N fractions clearly indicated that supplementation of *A. sowa* may improve the availability of nitrogen for microbial protein synthesis.

Supplementation of *A. sowa* increased (P<0.01) the production of total and individual VFAs in comparison to CON (Table 2). Chaves *et al.* (2008a) supplemented the diet with either carvacrol or cinnamaldehyde at 200 mg/kg DM intake in lambs and observed increase in total VFA concentration, confirming the earlier report of Castillejos *et al.* (2006). However, Chaves *et al.* (2008b) did not observe any effect of cinnamaldehyde (CDH), garlic and juniper berry essential oils (200 mg/kg of DM) on ruminal pH and concentrations of ammonia and total VFA. But Lin *et al.* (2013) reported that mixture of clove, oregano, cinnamon, and lemon or that of eugenol, carvacrol, citral and cinnamaldehyde supplemented at 1g/day in the diet of sheep decreased the VFA concentration. The acetate to propionate ratio improved (P<0.01) in the HFA group supplemented with *A. sowa* as compared to CON, confirming the reports of Lin *et al.* (2013) and Ma *et al.* (2015) in sheep. Supplementation of the TMR with *A. sowa* increased (P<0.01) the fermentation efficiency (E) and efficiency of conversion of fermented hexose energy to VFA energy (E₁) with a reduced (P<0.01) efficiency of conversion of hexose to methane (E₂). The relative proportion of acetate was depressed (P<0.01) in the HFA group, while that of iso-butyrate, butyrate and iso-valerate increased (P<0.01). Vakili *et al.* (2013) also observed similar trend in relative proportion of acetate and propionate in calves fed diet supplemented with

thyme and cinnamon essential oils. The relative proportion of branched chain fatty acids (BCFAs) was found higher in rumen of animals fed *A. sowa* in comparison to the animals fed the CON diet. The rumen profile clearly indicated that supplementation of diet with *A. sowa* improved the availability of nutrients for microbial protein synthesis, the ultimate substrate for the animal.

Table 2. Effect of supplementation of *A. sowa* on the volatile fatty acids in the rumen of Beetal bucks

Parameter	Dietary groups [†]		Pooled SE	P value
	CON	HFA		
TVFA, mM/dl	5.90 ^a	7.90 ^b	0.24	< 0.001
Acetate, mM/dl	4.06 ^a	4.95 ^b	0.16	0.001
Propionate, mM/dl	0.96 ^a	1.35 ^b	0.07	< 0.001
Isobutyrate, mM/dl	0.033 ^a	0.063 ^b	0.004	< 0.001
Butyrate, mM/dl	0.70 ^a	1.38 ^b	0.08	< 0.001
Isovalerate, mM/dl	0.05 ^a	0.09 ^b	0.005	0.001
Valerate, mM/dl	0.09	0.08	0.010	0.308
A:P ratio	4.23 ^b	3.67 ^a	0.010	0.003
<i>Relative proportion</i>				
Acetate, %	68.81 ^b	62.66 ^a	1.35	0.004
Propionate, %	16.27	17.09	0.63	0.084
Isobutyrate, %	0.56 ^a	0.80 ^b	0.03	< 0.001
Butyrate, %	11.86 ^a	17.47 ^b	0.78	0.002
Isovalerate, %	0.85 ^a	1.14 ^b	0.05	0.021
Valerate, %	1.52	1.01	0.14	0.046
<i>Fermentation efficiency[‡], %</i>				
E	73.24 ^a	74.82 ^b	0.30	0.003
E ₁	73.01 ^a	74.54 ^b	0.27	0.001
E ₂	17.12 ^b	15.67 ^a	0.26	< 0.001

[†]Feeding of basal diet in form of total-mixed ration alone (CON) or supplemented with *A. sowa* at 2% level (HFA).

[‡]E: Efficiency of rumen fermentation, E₁: Efficiency of fermented hexose energy to VFA energy, E₂: Efficiency of fermented hexose to methane

^{ab}Means with different superscripts in a row differ significantly (P < 0.05)

Nutrient utilization

The daily DM intake from diet supplemented with HFA was statistically comparable to CON (Table 3). Similar results have been reported earlier also, when the diet of lamb or sheep was supplemented with essential oils either individually or in different combinations (Chaves *et al.*, 2008a,b; Klevenhusen *et al.*, 2011; Lin *et al.*, 2013; Ma *et al.*, 2015). Supplementation of *A. sowa* to the CON did not have any significant impact on the digestibility of nutrients (Table 3). A number of reports have confirmed that essential oil/blend of essential oils did not affect the digestibility of nutrients (Sallam *et al.*, 2009; Wang *et al.*, 2009; Santos *et al.*, 2010; Ahmed *et*

al., 2014). But *in vitro* (Bakshi *et al.*, 2004)/ *in vivo* (Bakshi and Wadhwa, 2004) studies revealed that nine herbal feed additives (*A. sowa* was one of the HFA) supplemented individually or in combination at 0.4% of the substrate, improved the digestibility of nutrients in buffalo calves, mainly due to increased activities of fiber degrading enzymes (Wadhwa and Bakshi, 2006). The nitrogen balance study revealed that nitrogen intake, N-balance and apparent biological value were comparable between *A. sowa* supplemented (Table 3) and control group, confirming the earlier report of Ahmed *et al.* (2014).

Table 3. Effect of supplementation of *A. sowa* on the dry matter intake, digestibility of nutrients and N-retention in Beetal kids

Parameter	Dietary groups [†]		Pooled SE	P value
	CON	HFA		
<i>Dry matter intake, g/d</i>				
Concentrate	290.95	316.76	7.94	0.105
Green	218.32	239.22	8.03	0.216
Total DM	509.27	555.98	19.66	0.545
<i>Digestibility of nutrients, %</i>				
DM	70.88	73.86	1.25	0.716
OM	74.22	76.12	1.17	0.674
CP	71.57	73.78	1.66	0.548
NDF	61.88	64.96	1.91	0.652
ADF	44.83	45.22	2.09	0.934
Cellulose	62.49	64.08	1.93	0.924
Hemicellulose	74.99	76.18	2.06	0.396
<i>N balance, g/d</i>				
N intake	13.69	14.23	0.53	0.664
Faecal N	3.88	3.64	0.12	0.490
Digestible N	9.81	10.58	0.59	0.570
Urinary N	2.60	3.02	0.42	0.370
N retention	7.21	7.56	0.24	0.954
ABV %	73.50	71.46	2.63	0.356

[†]Feeding of basal diet in form of a total-mixed ration alone (CON) or supplemented with *A. sowa* at 2% level (HFA)

The blood biochemical profile of the animals indicated that *A. sowa* had no adverse effects on the serum values of different constituents (Table 4). Earlier studies also revealed that plasma concentrations of glucose, cholesterol, triglyceride, urea-N, β -hydroxybutyrate, alanine amino transferase and aspartate amino transferase were not changed by supplementing the diet with thyme and cinnamon essential oils in calves (Vakili *et al.*, 2013). The excretion of purine derivatives in urine has been proposed as a good indicator of the synthesis of microbial protein in the rumen. The excretion of purine derivatives has been used to predict the amount of absorbed purine

bases in the small intestine, which is related to the synthesis of microbial protein (Balcells *et al.*, 1991; Chen and Gomes, 1992). The results revealed that on supplementation of diet with *A. sowa*, the urinary excretion of uric acid, allantoin and, in turn, the total purine derivatives in *A. sowa* supplemented diet group were statistically comparable to CON (Table 5). Allantoin constituted 96-97% of total PD excretion; the values were quite comparable to those reported earlier for sheep and goats (Lindberg, 1985; Dapoza *et al.*, 1999). The microbial protein synthesis was also comparable in both the groups.

Table 4. Effect of supplementation of *A. sowa* on the blood biochemical profile of Beetal kids

Parameter	Dietary groups [†]		Pooled SE	P value
	CON	HFA		
Glucose, mg/dl	59.50 ^a	79.0 ^b	5.69	0.011
Total protein, g/dl	5.99	6.09	0.24	0.884
Albumin, g/dl	2.54	2.64	0.10	0.729
Globulin, g/dl	3.43	3.46	0.14	0.881
A:G ratio	0.74	0.76	0.17	0.515
Urea, mg/dl	49.56	54.08	3.86	0.662
Uric acid, mg/dl	0.55	0.66	0.04	0.188
Creatinine, mg/dl	0.96	1.02	0.05	0.607
Total cholesterol, mg/dl	65.90	53.95	11.05	0.688
Calcium, mg/dl	9.7	9.60	0.46	0.937
Phosphorus, mg/dl	6.70	6.30	0.54	0.784

[†]Feeding of basal diet in form of total-mixed ration alone (CON) or supplemented with *A. sowa* at 2% level (HFA).

^{ab}Means with different superscripts in a row differ significantly ($P < 0.05$).

Table 5. Effect of supplementation of *A. sowa* on the urinary excretion of purine derivatives in Beetal kids

Parameter	Dietary groups [†]		Pooled SE	P value
	CON	HFA		
<i>Purine derivatives excretion, mM/W^{0.75}/d</i>				
Uric acid	0.006	0.007	0.001	0.514
Allantoin	0.221	0.248	0.022	0.690
Total	0.227	0.255	0.023	0.682
<i>Purine derivatives excretion, %</i>				
Allantoin	96.97	97.28	0.42	0.776
Urinary allantoin	3.03	2.72	0.42	0.776
Microbial N synthesis, g/d	0.38	0.44	0.25	0.921
Purine nitrogen index	0.061	0.057	0.005	0.760
Creatinine	0.074	0.087	0.101	0.236

[†]Feeding of basal diet in form of total-mixed ration alone (CON) or supplemented with *A. sowa* at 2% level (HFA)

Carcass parameters

The initial body weight of goat kids fed CON and HFA was 13.8 and 15.9 ± 0.53 kg and the final body weight after 90 days feeding trial was 16.72 and 19.48 ± 0.91 kg, resulting in average daily gain of 34.07 and 41.17 ± 10.54 g, respectively. The average daily gain in weight of kids fed CON or diet supplemented with *A. sowa* was statistically comparable. Similar results were reported by Bampidis *et al.* (2005) with supplementation of oregano leaves in the diet of growing lambs.

Supplementation of *A. sowa* in the ration of Beetal kids affected various carcass traits significantly ($P \leq 0.05$); however, carcass weight and dressing percentage in *A. sowa* supplemented group were comparable to those offered control diet (Table 6). The values are in confirmation with those reported for Angora whether goat (McGregor, 2000). Edible offal percentage in animals fed *A. sowa* was also statistically comparable in both the groups. The ready-to-cook carcass expressed on weight basis ($P < 0.05$) or as percent of live weight basis was improved significantly ($P < 0.01$) in *A. sowa* supplemented group as compared to those fed control diet.

The weight of almost all of the non-carcass components (head, feet, skin, heart, lungs etc.) was comparable, except that of tail and testis which were improved significantly ($P \leq 0.05$) in the group of animals fed diet supplemented with *A. sowa* as compared to un-supplemented control diet (Table 7), which might be due to higher pre-slaughter weight in *A. sowa* fed group.

The participation of cuts in the carcass allows a qualitative assessment because it should present the best possible proportion of cuts with higher content of edible tissues, mainly muscles or the best proportion of cuts of interest to consumers (Yanez *et al.*, 2006). Supplementation of *A. sowa* in the ration of *Beetal* kids improved the weight of neck, rack, fore saddle and hind saddle significantly ($P < 0.05$), while that

Table 6. Effect of supplementation of *A. sowa* on the carcass characteristics of Beetal kids

Parameter	Dietary groups [†]		Pooled SE	P value
	CON	HFA		
Slaughter weight, kg	15.52	18.32	0.90	0.126
Dressed carcass, kg	6.48	7.86	0.40	0.083
% LW	38.80	40.31	0.45	0.114
Edible offals, kg	0.732	0.917	0.08	0.149
% LW	4.38	4.71	0.13	0.223
Inedible offals, kg	3.41	3.81	0.28	0.340
% LW	20.40	19.56	0.29	0.139
Ready-to-cook carcass, kg	6.25 ^a	7.73 ^b	0.82	0.016
% LW	37.38 ^a	39.64 ^b	0.12	0.024

[†]Feeding of basal diet in form of total-mixed ration alone (CON) or supplemented with *A. sowa* at 2% level (HFA)

^{ab}Means with different superscripts in a row differ significantly ($P < 0.05$).

of loin-ramp, fore-shank and legs non-significantly, but reverse trend was observed in case of brisket (Table 8). Almost similar trend was observed when these primal cuts were expressed as percent of live weight. Overall it showed positive effect for *A. sowa* supplementation on primal cuts from goat kids which will be translated into higher commercial value in meat industry.

Table 7. Effect of dietary supplementation of *A. sowa* on non-carass components of Beetal kids

Parameter	Weight, g		Pooled SE	P value	Weight, % LW		Pooled SE	P value
	CON [†]	HFA			CON [†]	HFA		
Head	1308.6	1435.1	84.21	0.496	7.80	7.32	0.16	0.141
Feet	716.80	781.49	43.15	0.497	4.27	4.00	0.08	0.116
Tail	22.87 ^a	38.93 ^b	4.14	0.038	0.14 ^a	0.20 ^b	0.01	0.011
Skin	1291.8	1592.6	104.80	0.165	7.68	8.14	0.19	0.247
Heart	72.78	98.84	7.72	0.335	0.49	0.50	0.02	0.769
Lungs	166.49	212.0	18.92	0.258	0.99	1.07	0.05	0.427
Kidney	77.19	96.58	8.88	0.310	0.46	0.49	0.02	0.569
Spleen	43.92	58.61	6.02	0.249	0.26	0.30	0.03	0.373
Lungs-gall bladder	307.44	405.13	29.67	0.100	1.83	2.06	0.07	0.068
Intestine	1557.2	1861.7	157.87	0.174	9.31	9.52	0.15	0.525
Testis	45.28 ^a	96.87 ^b	12.10	0.016	0.27 ^a	0.49 ^b	0.05	0.002

[†]Feeding of basal diet in form of total-mixed ration alone (CON) or supplemented with *A. sowa* at 2% level (HFA)

^{ab}Means with different superscripts in a row, within a particular criterion, differ significantly (P<0.05).

Table 8. Effect of dietary supplementation of *A. sowa* on weight and proportion of primal cuts from the carcass of Beetal kids

Parameter	Weight, g		Pooled SE	P value	Weight, % LW		Pooled SE	P value
	CON [†]	HFA			CON [†]	HFA		
Neck	415.0 ^a	818.4 ^b	87.01	0.004	6.38 ^a	10.36 ^b	0.78	<0.001
Brisket	727.42	583.23	46.91	0.131	11.19 ^b	7.36 ^a	0.74	<0.001
Rack	825.42 ^a	1140.10 ^b	77.48	0.026	12.70 ^a	14.50 ^b	0.37	0.001
Loin-Ramp	1441.9	1790.20	107.31	0.106	22.20	22.72	0.25	0.347
Fore shank	1536.30	1765.6	89.04	0.221	23.72	22.42	0.32	0.025
Legs	1481.7	1796.5	100.39	0.122	22.84	22.81	0.16	0.944
Fore saddle, kg	3.11 ^a	4.42 ^b	0.45	0.023	18.59 ^a	22.63 ^b	0.79	<0.001
Hind saddle, kg	2.63 ^a	3.58 ^b	0.35	0.036	15.71 ^a	18.35 ^b	0.53	<0.001

[†]Feeding of basal diet in form of total-mixed ration alone (CON) or supplemented with *A. sowa* at 2% level (HFA)

The colour of meat is a foremost attribute for the acceptance of the meat by the consumer. Supplementing the TMR with *A. sowa* resulted in light (L^*) yellow (b^*) colour of meat as compared to the CON (Table 9). However the redness (a^*) of meat was not affected by *A. sowa* supplementation. The diet supplemented with *A. sowa* produced higher meat yield (P<0.05) with higher (P<0.01) fat but lower protein content as compared to control diet.

Table 9. Effect of dietary supplementation of *A. sowa* on the colour and chemical composition of meat of Beetal kids

Parameter	Dietary groups [†]		Pooled SE	P value
	CON	HFA		
<i>Meat colour (Hunter color values)</i>				
Lightness	24.55 ^b	18.43 ^a	1.46	0.025
Redness	6.53	7.29	0.34	0.30
Yellowness	8.25 ^b	6.78 ^a	0.35	0.023
<i>Meat composition[‡], %</i>				
DM	22.6 ^a	24.57 ^b	0.45	0.001
Nitrogen [§]	13.01 ^b	11.99 ^a	0.23	<0.001
Ether extract [§]	8.13 ^a	17.03 ^b	1.99	<0.001

[†]Feeding of basal diet in form of total-mixed ration alone (CON) or supplemented with *A. sowa* at 2% level (HFA)

[‡]Based on ready to cook carcass.

[§]On DM basis.

^{ab}Means with different superscripts in a row differ significantly (P<0.05).

CONCLUSION

Supplementation of TMR with *A. sowa* at 2% of DM intake increased fermentation efficiency resulting in significantly higher VFA production. The ready-to-cook carcass yield was improved significantly with dietary supplementation of *A. sowa* in the diet of the kids.

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